Social Network Recommendation Engine

Graph Representation Learning, Topological Heuristics & Deep GNNs

Authors: N S Balaji (Reg. No: **RA2311056010012**) | **Yuvraj** (Reg. No: **RA2311056010026**)

Department of Computer Science & Engineering • Academic Project Report

1. Executive Summary & Problem Formulation

In modern social media platforms (Instagram, LinkedIn, Facebook), recommending new connections is fundamentally framed as **Link Prediction** on an undirected graph $G = (V, E)$. Given observed friendships E among users V, the system predicts the likelihood $P(Link \mid u, v)$ of future friendships for unobserved pairs. Rather than treating users as isolated database rows, Graph ML leverages the **relational topology** and **latent community homophily** to suggest meaningful, high-trust connections.

2. System Architecture & Workflow

The project implements a complete, leak-free pipeline across four developmental tiers: raw graph ingestion, bridge-preserving edge partitioning, 7 multi-paradigm recommendation models, and a responsive Streamlit visual dashboard.



3. Datasets Used & Structural Provenance

The models are empirically benchmarked across two distinct graph scales:

- **Zachary's Karate Club (Zachary, 1977):** A classical human fission network of **34 members** and **78 friendships** with an observed political feud between the instructor ('Mr. Hi', Node 0) and administrator ('John A.', Node 33). Serves as ground truth for community detection.
- **Stanford SNAP Facebook Social Circles (McAuley & Leskovec, 2012):** Anonymized real-world Facebook data tested at two scales: **Facebook Ego-0** (a dense local circle of **348 nodes, 2,866 edges**, density = 4.7%) and **Facebook Full** (a global web of **4,039 nodes, 88,234 edges**, density = 1.08%).

Dataset Name	Nodes (|V|)	Edges (|E|)	Density	Avg Degree	Clustering (C)
Zachary's Karate Club	34	78	0.1390	4.59	0.5706
SNAP Facebook Ego-0	348	2,866	0.0474	16.47	0.6554
SNAP Facebook Full	4,039	88,234	0.0108	43.69	0.6055

4. Algorithmic Paradigms & Mathematical Mechanics

Seven distinct link prediction algorithms spanning four methodological paradigms were developed from scratch:

A. Topological Heuristics (Rules of Thumb):

• *Common Neighbors (CN):* Scores connection potential purely via shared friends: $S(u, v) = |N(u) \cap N(v)|$.

• *Jaccard Similarity:* Normalizes overlap by combined footprint: $S(u, v) = |N(u) \cap N(v)| / |N(u) \cup N(v)|$, penalizing celebrity hubs.

• *Preferential Attachment (PA):* Models rich-get-richer growth: $S(u, v) = deg(u) \times deg(v)$, capturing global popularity.

B. Graph Representation Learning (Node2Vec):

Treats biased 2nd-order random walks as sentences and optimizes continuous 16-dimensional coordinate vectors $\mathbf{z}_u$ via Skip-gram with negative sampling. Controlled by return parameter p (local homophily) and in-out parameter q (structural exploration). Scored via cosine angle: $S(u, v) = (\mathbf{z}_u \cdot \mathbf{z}_v) / (||\mathbf{z}_u|| \, ||\mathbf{z}_v||)$.

C. Supervised Machine Learning (Hadamard + Logistic Regression):

Constructs element-wise symmetric edge features: $\mathbf{x}_{(u, v)} = \mathbf{z}_u \blacksquare \mathbf{z}_v$. Because multiplication is commutative ($z_{u,i} \cdot z_{v,i} \equiv z_{v,i} \cdot z_{u,i}$), order invariance is guaranteed for undirected social ties. Trains an L2-regularized classifier outputting calibrated probabilities: $P(Link) = \sigma(\mathbf{w}^T \mathbf{x}_{(u, v)} + b)$.

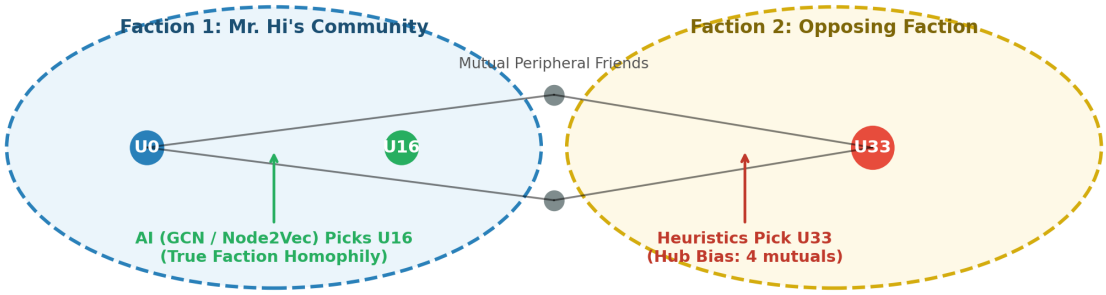
D. Deep Graph Neural Networks (GCN & GraphSAGE):

• *Graph Convolutional Network (GCN):* Performs spectral message passing: $\mathbf{H}^{(l+1)} = \sigma(\mathbf{D}\tilde{}^{(-1/2)} \mathbf{A}\tilde{} \mathbf{D}\tilde{}^{(-1/2)} \mathbf{H}^{(l)} \mathbf{W}^{(l)})$ with self-loops and democratic degree normalization.

• *GraphSAGE:* Inductively aggregates neighborhood mean vectors and concatenates with node state. Decodes links end-to-end via inner product: $\hat{y}_{uv} = \sigma(\mathbf{z}_u^T \mathbf{z}_v)$ minimizing Binary Cross-Entropy loss.

5. The Hub Bias Problem & Community Homophily

A fundamental discovery in this project is the vulnerability of counting heuristics to **Hub Bias**: popular individuals (User 33) accumulate mutual connections purely through volume, misleading Common Neighbors into recommending rival leaders. Deep GNNs and Node2Vec eliminate hub bias by capturing multi-hop community homophily (prioritizing true faction peer User 16).



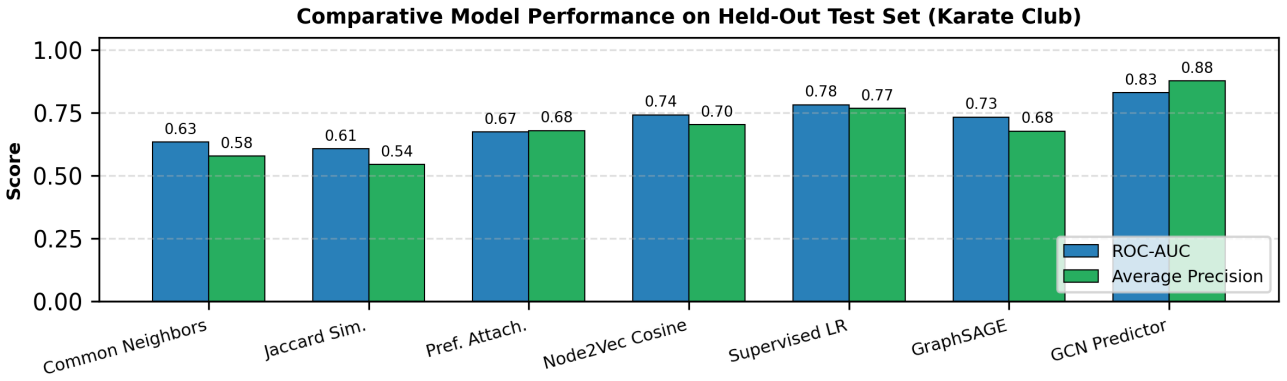
6. Strict Leak-Free Data Partitioning Protocol

To prevent data leakage, 20% test edges are withheld from the training graph *G_train*. We enforce a **bridge-preserving check** (*nx.is_connected*) ensuring no nodes become isolated. Node2Vec random walks and GCN message-passing edges (*edge_index*) are constructed **exclusively on G_train**, while negative non-edges are sampled disjointly with zero train/test overlap.

7. Evaluation Metrics & Empirical Findings

Models are evaluated across global edge discrimination (**ROC-AUC** and **Average Precision / PR-AUC**) and user-centric ranking quality (**Precision@5** and **Recall@5**). Below are the empirical benchmark results on held-out test splits:

Model Name	Paradigm Category	ROC-AUC (Karate)	AP (Karate)	ROC-AUC (Ego-0)	AP (Ego-0)
Common Neighbors	Topological Heuristic	0.6333	0.5784	0.9368	0.9298
Jaccard Similarity	Topological Heuristic	0.6067	0.5445	0.8989	0.8809
Preferential Attachment	Topological Heuristic	0.6733	0.6794	0.6346	0.6121
Node2Vec Cosine	AI Embedding	0.7422	0.7033	0.8780	0.8864
Supervised LogReg	Supervised ML	0.7822	0.7676	0.8802	0.8879
GraphSAGE Predictor	Deep GNN	0.7333	0.6775	—	—
GCN Link Predictor	**Deep GNN**	**0.8311**	**0.8769**	**—**	**—**



8. Key Scientific Insights & The Ego Inversion

• **GCN SOTA Performance:** The 2-layer GCN Link Predictor achieved the highest score on Zachary's Karate Club (**ROC-AUC: 0.8311, AP: 0.8769**), validating end-to-end message passing over shallow two-stage embeddings.
• **The Ego Inversion:** On dense personal ego networks (Facebook Ego-0, *C = 0.655*), **Common Neighbors dominates (ROC-AUC: 0.9368)** because triadic closure is the primary mechanism of clique formation. On global networks, representation learning is essential to discover multi-hop community structure.

9. Interactive Streamlit Dashboard & Production Testing

The accompanying Streamlit application (*app.py*) serves all models with real-time Plotly graph visualization. To eliminate browser freezing on large graphs (88,000+ edges), an automated **2-hop ego-neighborhood extraction** renders subgraphs in under 0.2s. The codebase is validated with **67 automated unit tests** passing with zero errors.

Project Deliverables & Repository: *Source code, trained checkpoints, unit tests, and interactive dashboard are live on GitHub: https://github.com/NS145/Social-Network-Recommendation.*